

**Learned Word Representation.** Word translation is a popular intrinsic evaluation task for cross-lingual word representations. Therefore, we evaluate the word representations learned by the BI-LSTM embedder on a word translation benchmark. Specifically, we use the SRC embedder to generate embeddings for all French, Italian, and Spanish words that appear in multiCCA’s vocabulary and translate each word with nearest-neighbor search. Table 6 shows the top-1 word translation accuracy on the test dictionaries from MUSE (?). Although the SRC embedder is not exposed to any cross-lingual signal, it rivals CLWE on the word translation task by exploiting character-level similarities between languages.

**Qualitative Analysis.** To understand how cross-lingual character-level similarity helps classification, we manually compare the output of a CLWE-based model and a CACO model (DICT variant) from the Spanish to Italian CLDC experiment. Sometimes CACO avoids the mistakes of CLWE-based models by correctly aligning word pairs that are misaligned in the pre-trained CLWE. For example, in the CLWE, “relevancia” (relevance) is the closest Spanish word for the Italian word “interesse” (interest), while the CACO embedder maps both the Italian word “interesse” (interest) and the Spanish word “interesse” (interest) to the same point. Consequently, CACO correctly classifies an Italian document about the interest rate with GCAT (government), while the CLWE-based model predicts MCAT (market).

## 4 Related Work

Previous CLDC methods are typically word-based and rely on one of the following cross-lingual signals to transfer knowledge: large bilingual lexicons (?; ?), MT systems (?; ?; ?), or CLWE (?). One exception is the recently proposed multilingual BERT (?; ?), which uses a subword vocabulary. Unfortunately, some languages do not have these resources. CACO can help bridge the resource gap. By exploiting character-level similarities between related languages, CACO can work effectively with few or no target language data.

To adapt CLWE to low-resource settings, recent unsupervised CLWE methods (?; ?) do not use dictionary or parallel text. These methods can be further improved with careful normalization (?) and interactive refinement (?). However, unsupervised CLWE methods still require large monolingual corpora in the target language, and they might fail when the monolingual corpora of the two languages come from different domains (?; ?) and when the two language have different morphology (?). In contrast, CACO does not require any target language data.

Cross-lingual transfer at character-level is successfully used in low-resource paradigm completion (?), morphological tagging (?), part-of-speech tagging (?), and named entity recognition (?; ?; ?; ?), where the authors train a character-level model jointly on a small labeled corpus in target language and a large labeled corpus in source language. Our method is similar in spirit, but we focus on CLDC, where it is less obvious if orthographic features are helpful. Moreover,

| source                                                                                                                                                                    | target                                                                                 | SRC  | DICT | MIM  | ALL  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------|------|------|------|
|   FR/IT |  ES | 58.8 | 67.0 | 55.8 | 65.3 |
|   ES/IT |  FR | 51.8 | 55.8 | 50.3 | 56.0 |
|   ES/FR |  IT | 53.2 | 56.1 | 55.9 | 56.5 |
| average                                                                                                                                                                   |                                                                                        | 54.6 | 59.6 | 54.0 | 59.3 |

Table 5: Results of CLDC experiments using two source languages. Models trained on two source languages are generally better than models trained on only one source language (Table 2).

| source                                                                               | target                                                                                 | CLWE | CACO |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------|------|
|  ES |  FR | 36.8 | 31.1 |
|  ES |  IT | 44.0 | 33.1 |
|  FR |  ES | 34.0 | 30.9 |
|  FR |  IT | 33.5 | 29.6 |
|  IT |  ES | 42.1 | 37.5 |
|  IT |  FR | 35.6 | 36.4 |
| average                                                                              |                                                                                        | 37.7 | 33.1 |

Table 6: Word translation accuracies (P@1) for different embeddings. The CACO embeddings are generated by the embedder of a SRC model trained on the source language. Without any cross-lingual signal, the CACO embedder has competitive word translation accuracy as CLWE pre-trained on large target language corpora and dictionaries.

we introduce a novel multi-task objective to use different types of monolingual and cross-lingual resources.

## 5 Conclusion

We investigate character-level knowledge transfer between related languages for CLDC. Our transfer learning scheme, CACO, exploits character-level similarities between related languages through shared character representations to generalize from source language data. Empirical evaluation on multiple related language pairs confirm that character-level knowledge transfer is highly effective.

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